##

Sub-Threshold Jitter Governor - FPGA Timing & Physical Constraints

Target Frequency: 100 MHz (10.000 ns clock period)

Substrate Target: Intel Loihi 2 Neurocore / Custom FPGA RTL

##

1. Primary Clock Definition (100 MHz system clock on pin clk)

create_clock -name clk -period 10.000 [get_ports clk]

Set clock uncertainty for jitter modeling (matching 0.46 ms AER temporal jitter envelope scaled to cycle domain)

set_clock_uncertainty 0.120 [get_clocks clk]

2. Input and Output Delay Constraints relative to clk

Input constraints for asynchronous jitter measurements and enable signals

set_input_delay -clock clk -max 2.500 [get_ports {measured_jitter_us[*] enable rst_n}]
set_input_delay -clock clk -min 0.500 [get_ports {measured_jitter_us[*] enable rst_n}]

Output constraints for adaptive leak coefficients and delay-line offsets

set_output_delay -clock clk -max 1.800 [get_ports {leak_coefficient[*] delay_line_offset_us[*] jitter_locked}] set_output_delay -clock clk -min 0.300 [get_ports {leak_coefficient[*]

*delay_line_offset_us[*] jitter_locked}]

3. False Paths & Asynchronous Control Crossings

Asynchronous reset path

set_false_path -from [get_ports rst_n]

4. Multicycle Paths for Adaptive PI Integrator Feedback Loop

The lock_counter and adaptive leak registers update on slower control loops (multicycle 2)

set_multicycle_path 2 -setup -from [get_cells {leak_coefficient_reg[*]}] *-to [get_cells {lock_counter_reg[*]*}] set_multicycle_path 1 -hold -from [get_cells {leak_coefficient_reg[*]}] *-to [get_cells {lock_counter_reg[*]*}]

##

Xilinx XDC Physical & Environmental Constraints (Optional for Vivado)

##

set_property PACKAGE_PIN E3 [get_ports clk]

set_property IOSTANDARD LVCMOS33 [get_ports clk]

set_property BITSTREAM.CONFIG.CONFIGRATE 50 [current_design]